

Kyle Mogan

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SOFTWARE ENGINEERING SUMMARY

Senior Software Engineer with 10+ years of experience building and modernizing **large-scale healthcare platforms** at a **Fortune 5 enterprise**. Experienced in developing full-stack and data-driven applications using modern frameworks, AI concepts, and cloud-native technologies. Specialized in **high-throughput backend systems, cloud modernization, and data-intensive processing**, delivering measurable gains in **performance, reliability, security, cost efficiency, and innovation**.

- Designed systems processing **1.2M+ members per run in minutes** and **hundreds of thousands of records in seconds**
- Led modernization initiatives resulting in **\$3M+ annual cost savings**
- Improved API and batch performance by **95%+** across critical healthcare workflows
- Extensive experience across **monolith and microservice architectures**, cloud, CI/CD, and data streaming platforms

SKILLS SUMMARY

Back End Development: Java • Kotlin • Python • C# • C • C++ • PHP • Spring Boot / Batch • Node.js

Front End & Mobile Development: HTML • CSS • JavaScript • AJAX • Flutter • Dart

Databases: MySQL • PostgreSQL • MongoDB • Cloud Firestore

Cloud & DevOps: GCP • AWS • Azure • Kubernetes • Docker • GitHub Actions • Jenkins • CI/CD

Data & Streaming Platforms: Kafka • Pub/Sub • Dataflow • Databricks

AI/ML: Natural Language Processing • ChatGPT API Integration • Auto ML • BigQuery ML • LangGraph • ChromaDB • RAG

Monitoring & Analytics: Splunk • Google Analytics

Tools & Practices: Agile • JSON • XML • Git • Maven • Gradle • SonarQube • SonarLint • Spock

Operating Systems & Environments: macOS • Linux • Windows • Android • Computer Hardware

EMPLOYMENT HISTORY

Senior Software Engineer • February 2022 – Present

Optum is a **global healthcare technology** and services organization supporting payers, providers, and governments, operating on a national scale across millions of members. Optum is a **Fortune 5** company with **hundreds of billions of dollars in total revenue** across all lines of business.

Senior software engineer on a **newly formed modernization team** operating with **startup-like autonomy** to **redesign** and **externalize legacy healthcare processing systems**.

Optum, Minnetonka, MN

- Designed and implemented **microservices** for healthcare systems, including capitated services, using **Java, Spring Boot/Batch, Kafka**, and **Databricks** for real-time data streaming and analytics
- Engineered a capitation engine capable of processing **1.2 million members in 42 minutes**, significantly reducing total runtime
- Collaborated with a **globally distributed medium team (~30 people)** to manage developers, enforce coding standards, and support **Agile** and **Scrum practices**; **reduce story splits by 40%** through accurate sprint planning
- Built a high-performance file generation utility capable of processing **372,000 members in 22 seconds**
- Migrated services from on-prem to **cloud** and **hybrid-cloud**, resulting in **\$3 million in savings**
- Led **externalization** of **legacy monolith components**, extracting them into **independently deployable services**
- Led **modernization initiatives**, converting legacy monolith components into **new microservices** to enhance **performance, scalability, and maintainability**
- Identified and resolved **performance bottlenecks**, improving API and job throughput by **95%+** via caching, multithreading, and process simplification
- Innovated **AI-driven solutions**, leveraging ChatGPT and other ML tools to improve development quality, automation, and insights
- Integrated services into CI/CD pipelines (**GitHub Actions, Jenkins**) and deploy across **Kubernetes, Azure**, and **Google Cloud Platform (GCP)** environments
- Led **database optimization and migration** projects: MySQL → PostgreSQL and Cassandra → MongoDB
- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline using **Python, LangGraph, ChromaDB, Azure Databricks**, and **GPT-4 models** to support question-answering over medical documentation
- Engineered **validation pipelines, embedding strategies**, and **human-in-the-loop** review mechanisms to ensure correctness, compliance, and safe deployment of AI-assisted outputs in regulated environments

AI Systems Engineer • January 2025 – Present

AI Systems Engineer responsible for designing and implementing **LLM-powered natural language interfaces** and **Retrieval-Augmented Generation (RAG) pipelines** within a production CRM platform, enabling secure, validated database query generation and **AI-assisted workflow automation**.

RecordBookHQ, Remote – Part-Time Contract

- Architected **LLM-powered natural language query interface** translating structured model outputs into validated dynamic database queries
- Implemented **schema-constrained JSON outputs** and **server-side validation** to ensure deterministic and secure execution
- Integrated **OpenAI APIs** into Laravel-based backend systems to support **AI-assisted CRM workflows**

Software Engineer • April 2017 – February 2022

Software engineer on a highly configurable and resilient **monolithic healthcare claims adjudication platform** supporting claims, members, providers, networks, and contracts.

Optum, Minnetonka, MN

- Developed and maintained **claims adjudication systems** supporting complex healthcare workflows
- Engineered an adjudication engine achieving a **95%+ auto-adjudication rate**
- Implemented **workflow-driven backend business logic** aligned with domain rules and processing pipelines
- Led development of **division of financial responsibility** and **delegation engines**
- Built and maintained **RESTful APIs** and **UI components** for internal and customer-facing systems
- Implemented **logging, monitoring, and code quality practices** using **Splunk, SonarQube**, and automated unit/integration testing (Java, Groovy)
- Performed **production triage** ensuring **99% uptime** and rapid issue resolution across multiple environments
- Built and deployed applications utilizing **Jenkins** pipelines
- Worked on a **large co-located team (~60 people)** of software engineers, system architects, and business and system analysts
- **Coordinated** with **several co-located teams** all working on the same product (**~250 people**)
- Coordinated with contractors and employees in **India**
- **Promoted to Senior Software Engineer** due to **technical expertise** and **high performance**

Senior Apps Developer • July 2016 – April 2017

Senior application developer on a **monolithic healthcare administration platform**; promoted rapidly based on performance and technical ramp-up speed.

Optum, Eden Prairie, MN

- Developed and maintained **UIs, automation tests, and unit tests** for a **monolithic healthcare administration platform** (handling claims, members, providers, networks, and contracts) that enables **increased speed** and **flexibility for new markets** and **reduced costs**
- Built applications using **MVC** patterns, utilizing **Java, MySQL, Javascript, CSS, and HTML**
- Contributed to **test automation frameworks** and **testing standards**
- Worked on a **medium co-located team (~35 people)** of software engineers, system architects, and business and system analysts
- **Rapidly onboarded** to internal frameworks and **delivered production features within weeks**
- **Promoted to Software Engineer** due to **impact** and **consistency**

Technology Development Program Senior Associate • June 2015 – July 2016

Senior associate on a rotational engineering program exposing multiple **healthcare platforms, teams, and architectures**.

Optum, Eden Prairie, MN

- Completed multiple **6-month rotations** across **legacy** and **modern healthcare systems**
- Developed **UI enhancements** and **unit tests** for systems processing **healthcare liability transactions** and **consumer accounts**
- Participated in **architecture bootcamps** and **internal technical training programs**
- Worked on a **small co-located team (~20 people)** of software engineers, system architects, and business and system analysts
- **Excelled** in onboarding **bootcamp curriculum**, completing ahead of peers
- Graduated early from the program and **promoted to Senior Apps Developer**

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

GOOGLE CLOUD CERTIFIED DIGITAL LEADER • 2024

Developed foundational expertise in Google Cloud technologies, including:

- **Digital Transformation:** Cloud adoption strategies, cost optimization (CapEx → OpEx), scalability, and hybrid/multicloud models
- **Data & AI:** Data warehousing (BigQuery, Cloud SQL), analytics pipelines (Pub/Sub, Dataflow), and AI/ML services (AutoML, BigQuery ML)
- **Infrastructure & Security:** Application modernization, containers, serverless computing, IAM, compliance, and operational reliability (SRE principles)
- **Cloud Governance:** Financial management, sustainability, and trust frameworks in cloud operations

EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

University of Minnesota, Minneapolis, MN

SYSTEMS AND LOW-LEVEL ENGINEERING

- **Unix Systems Programming (C):** Memory management, pointers, file descriptors, processes, signals, inter-process communication, and debugging with gdb
- **Operating Systems:** Threading models, synchronization primitives (mutexes, semaphores), scheduling, concurrency control, and performance analysis
- Developed and debugged low-level C applications in Linux environments, emphasizing **deterministic behavior, resource management, synchronization, and system-level reliability** in performance-sensitive contexts